WORKED SOLUTION

STEP III 2000, Question 10

Mechanics • Friction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q10

The key idea

Start with the geometry: the distances $\sqrt{3}\,a$ and a are chosen so that the beam meets the ground at a special angle, and that fixes every direction in the problem. Then take moments about the two points where unknown forces drop out. About the hinge X, the beam's equation contains only the normal reaction from the sphere, because the friction from the sphere acts along the beam itself. About the centre of the sphere, every force except the two frictions passes through the centre, so the two frictions must be equal.

Solution

Model and geometry

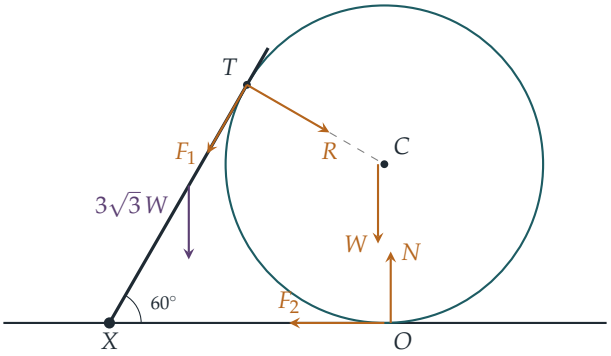
We treat the beam as a uniform rod and the sphere as a uniform rigid body whose weight acts at its centre C. Both contacts are rough. Let O be the point where the sphere touches the ground, so $OX = \sqrt{3}\,a$ and $OC = a$, with OC vertical.

The beam touches the sphere at a point T, and it is tangent to the sphere there. The ground is the other tangent from X. In the right-angled triangle XOC,

$$\tan\angle OXC = \frac{a}{\sqrt{3}\,a} = \frac{1}{\sqrt{3}}, \qquad \text{so} \qquad \angle OXC = 30^\circ.$$

The two tangents from X are symmetric about the line XC, so the beam makes an angle $2 \times 30^\circ = 60^\circ$ with the ground. Tangent lengths from a point are equal, so $XT = XO = \sqrt{3}\,a$. Since $\sqrt{3}\,a < 2a$, the contact point really does lie on the beam, a little beyond its midpoint.

The radius CT is perpendicular to the beam. So the normal reaction at T makes 30° with the horizontal, and the friction at T acts along the beam, at 60° to the horizontal.



The amber arrows are the forces on the sphere. The beam feels R and F_1 in the opposite directions, together with its weight $3\sqrt{3}\,W$ at its midpoint (a distance a from X) and an unknown reaction at the hinge.

We have guessed the directions of the frictions: on the sphere, F_1 acts down the beam towards X, and F_2 acts along the ground towards X. If a guess is wrong, that force will come out negative.

The beam: moments about X

The hinge reaction acts at X, so it has no moment about X. The friction from the sphere acts along the beam, and the beam's line passes through X, so it has no moment either. The normal reaction R is perpendicular to the beam at distance $XT = \sqrt{3}\,a$. The weight acts at horizontal distance $a\cos 60^\circ = \frac{1}{2}a$ from X. Hence

$$R \cdot \sqrt{3}\,a = 3\sqrt{3}\,W \cdot \tfrac{1}{2}a, \qquad \text{so} \qquad R = \tfrac{3}{2}W.$$

The sphere: moments about C

The forces R, W and N all pass through C. (The normal reactions act along radii.) The two frictions are tangential, each at distance a from C. F_1 at T turns the sphere one way about C and F_2 at O turns it the other way, so

$$F_1 a = F_2 a, \qquad \text{so} \qquad F_1 = F_2.$$

Call this common value F.

> **Ask yourself** Why can we be sure the two frictions are equal, before knowing anything else about them?

The sphere is in equilibrium, so the total moment about any point is zero. The centre is the clever choice: the weight and both normal reactions pass through it, so only the frictions are left, and they have equal lever arms a.

The sphere: resolving

Horizontally, taking the direction away from X as positive: R has component $R\cos 30^\circ$, F_1 has component $-F\cos 60^\circ$, and F_2 contributes $-F$. So

$$\frac{\sqrt{3}}{2}R = \frac{1}{2}F + F = \frac{3}{2}F, \qquad \text{so} \qquad F = \frac{R}{\sqrt{3}} = \frac{\sqrt{3}}{2}W.$$

This is positive, so the guessed friction directions were right. Vertically, R pushes down with component $R\sin 30^\circ$, and F_1 pulls down with component $F\sin 60^\circ$. So

$$N = W + \tfrac{1}{2}R + \tfrac{\sqrt{3}}{2}F = W + \tfrac{3}{4}W + \tfrac{3}{4}W = \tfrac{5}{2}W.$$

The coefficients of friction

The sphere is on the point of slipping at both contacts, so friction is limiting at both: $F_1 = \mu_1 R$ and $F_2 = \mu_2 N$. Therefore

$$\mu_1 = \frac{F}{R} = \frac{\frac{\sqrt{3}}{2}W}{\frac{3}{2}W} = \frac{1}{\sqrt{3}}, \qquad \mu_2 = \frac{F}{N} = \frac{\frac{\sqrt{3}}{2}W}{\frac{5}{2}W} = \frac{\sqrt{3}}{5}.$$

Checks

- A numerical check with coordinates ($a = 1.3$, $W = 2$) confirmed that T lies on the sphere with CT perpendicular to the beam. Solving the sphere's three equilibrium equations directly gave $R = 1.5W$, $F_1 = F_2 \approx 0.8660W$ and $N = 2.5W$, so $\mu_1 \approx 0.5774 = 1/\sqrt{3}$ and $\mu_2 \approx 0.3464 = \sqrt{3}/5$.
- An independent equation: take moments for the sphere about O. Now N, F_2 and W have no moment. The line of R passes through C, so its moment is that of its horizontal component at height a, namely $\frac{\sqrt{3}}{2}R\,a$. The moment of F_1 about O works out to $\frac{3}{2}aF$. Equating these gives $F = R/\sqrt{3}$ again, without using the moment equation about C.
- Both coefficients are less than 1, which is physically sensible for ordinary rough contacts.

Common slips

- Taking the beam's angle to be 30° (the angle of XC) instead of 60°.
- Forgetting the vertical component of F_1 when finding N. This gives $N = \frac{7}{4}W$ and the wrong μ_2.
- Including the friction from the sphere in the beam's moment equation about X. It acts along the beam, so its moment about X is zero.

Beyond STEP

A rigid body in a plane has three degrees of freedom, so it gives three independent equilibrium equations. Here the sphere has five unknown contact forces, and the beam supplies R to cut this down. The two "limiting friction" conditions then turn the problem into a solvable system. Counting equations against unknowns like this, and spotting when a system is statically indeterminate, is the starting point of structural mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully.

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